

Shanmukha Vishnu Vardan Melam

Gen AI Engineer | AI/ML Engineer

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SUMMARY

Aspiring Generative AI Engineer with hands-on experience in Machine Learning, Deep Learning, TensorFlow, Flask, REST APIs and end-to-end AI application development. Strong foundation in Python, supervised and unsupervised learning, data preprocessing, model evaluation, NLP fundamentals, Generative AI concepts, GPT-based models, prompt engineering, Familiar with TensorFlow, PyTorch and Hugging Face Transformers. Passionate about building production-ready AI solutions and continuously learning emerging AI technologies.

TECHNICAL SKILLS

Languages: Python, SQL

Generative AI: GPT, LLM Fundamentals, Prompt Engineering, Hugging Face Transformers, LangChain

ML/AI Frameworks: TensorFlow, Scikit-learn, CNN, Deep Learning

Backend: Flask, Django, REST API Development

Frontend: React.js, HTML5, CSS3

Databases: MongoDB, MySQL

Tools & Platforms: Git, GitHub

Core Concepts: Data Structures & Algorithms, OOPs, Supervised Learning

PROJECTS

Generative AI Chat Assistant

Jan 2026 - Apr 2026

Tech Stack: Python | OpenAI API | LangChain | Flask | GitHub: github.com/vishnumelam/genai-chat

- Developed an AI-powered conversational assistant using GPT-based Large Language Models (LLMs) to answer user queries with natural, context-aware responses.
- Implemented Retrieval-Augmented Generation (RAG) by integrating LangChain with FAISS vector database to retrieve relevant information from custom documents before generating responses.
- Built an end-to-end document ingestion pipeline supporting PDF and text files, including document loading, chunking, embedding generation, and semantic search.
- Utilized OpenAI Embeddings Hugging Face Sentence Transformers for vector embedding generation and efficient similarity search.

AI-Powered Tomato Disease Detection System |

Mar 2025 – May 2025

Tech Stack: Python, TensorFlow, CNN, Flask, React.js | GitHub: github.com/vishnumelam/tomato-disease-detection

- Designed and trained a Convolutional Neural Network (CNN) on the PlantVillage dataset for multi-class tomato leaf disease classification, delivering real-time predictions with confidence scores.
- Performed data preprocessing, image augmentation, and feature engineering to improve model generalization and reduce overfitting; achieved high validation accuracy across 10 disease classes.
- Built a Flask REST API to handle image uploads and serve model inference results; integrated with a React frontend supporting live image capture for a seamless end-user experience.

CERTIFICATIONS

Oracle Certified Foundations Associate - Agentic AI - Oracle University | Aug 2026

Retrieval-Augmented Generation for Enhanced AI Outputs — IBM SkillsBuild | Aug 2026

Introduction to RAG — CodeSignal | Aug 2026

EDUCATION

Lovely Professional University, Punjab ·

2021 – 2025

B.Tech, Computer Science and Engineering — CGPA: 7.61

Sri Chaitanya Junior College, Andhra Pradesh ·

2019 – 2021

Intermediate (Class XII) — 96.5%